

Grade 6 Accelerated
Unit 15
Lesson 4 Practice

Name _____
Date _____
Period _____

1. The probabilities of four events are shown. List them in order from least to greatest.

$$\frac{6}{13}$$

3 out of 9

41%

0.2

Use the spinner to determine missing events, probabilities, and descriptions of likelihood. Express all probabilities as percentages. Use the vocabulary to complete each sentence.

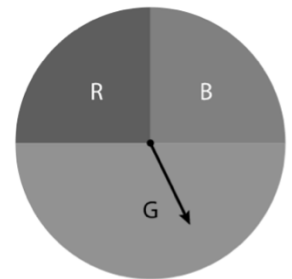
impossible

unlikely

equally likely as
unlikely

likely

certain



2. The probability of landing on green, $P(G)$, is _____. This event is _____.
3. The probability of landing on red, $P(R)$, is _____. This event is _____.
4. The probability of landing on yellow, $P(Y)$, is _____. This event is _____.
5. The probability of landing on red or blue, $P(R \text{ or } B)$, is _____. This event is _____.
6. The probability of landing on a color, $P(\text{color})$ is _____. This event is _____.

Here are some scenarios for questions 7-8.

- According to market research, a business has a 75% chance of making money in the first 3 years.
- According to lab testing, $\frac{5}{6}$ of a certain kind of experimental light bulb will work after 3 years.
- According to experts, the likelihood of a car needing major repairs in the first 3 years is 0.7.

7. Write the scenarios in order of likelihood from least to greatest after three years: the business makes money, the light bulb still works, and the car needs major repairs.
8. Name another chance experiment that has the same likelihood as one of the scenarios.
9. Which event is more likely: rolling a standard number cube and getting an even number, or flipping a coin and having it land heads up?
10. Miami International airport reported that on a given day, the probability that an airline arrived on time was 0.32. The likelihood that the next airplane arrived earlier than scheduled was $\frac{3}{10}$ and the later than scheduled was 37.8%. What is most likely to occur when the next airplane arrives?